Frontend Engineering II

FEE II Evaluation - 2 (Phase 2)

Semester-3 (Batch-2025)



Supervised By:

Mr. Shubham Kumar
(Technical Trainer)

Submitted By (Group G-15):

Kanav Chadha (Team Leader) - 2510990649
Kishan - 2510991457
Arsh Behl - 2510991354
Soumajit Majee - 2510990383

Department of Computer Science and Engineering
Chitkara University Institute of Engineering & Technology,
Chitkara University, Punjab

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PheW TuBe: Video Streaming Platform

Front End Engineering - II (25CSE0203) — Project Based Evaluation-2 (Batch-2025)

Department of Computer Science & Engineering • Chitkara University, Punjab

Supervised By:

Mr. Shubham Kumar

Technical Trainer, Department of CSE
Chitkara University Institute of Engineering & Technology

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Kanav Chadha — Roll No: 2510990649 (G-15)

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Evaluation-II Vision & Milestone Objectives

Phase 1 vs Phase 2 Focus

- **Phase 1 Foundation:** Established responsive video card grids, category chips, and external links.
- **Phase 2 Transition:** Elevated PheW TuBe into an authentic, stateful Single Page Application (SPA).
- **Zero Page Reloads:** Client-side routing via React Router DOM ensures instant state reactivity without full flushes.

Core Technical Objectives

- **In-Site Streaming:** Deep link /watch/:id powered by useParams() with embedded 16:9 streaming.
- **DOM Control via useRef:** Direct focus on comment inputs and auto-scroll control without synthetic re-renders.
- **Global Context API:** Centralized multi-state bus eliminating prop drilling across nested component trees.

User Experience & Realism

- **Up-Next Queue:** Dynamic video swapping via useNavigate() with synchronized channel metadata.
- **Working Action Controls:** Real Like increments, Share link copier, and Watch Later bookmarking.
- **Responsive Parity:** Fluid scaling across 375px mobile, 768px/1024px tablet, and 1440px desktop screens.

Syllabus-Aligned Technology Stack & Architecture

React 18 (Core SPA Engine)

- Declarative UI components.
- Virtual DOM fiber reconciliation.
- Concurrent state batching.
- **Hooks:** useState, useEffect, useRef.

React Router v6

- Client-side routing engine.
- Dynamic URL parameters (/watch/:id).
- Declarative <ProtectedRoute /> security.
- Seamless useNavigate() transitions.

Tailwind CSS 3

- Utility-first responsive framework.
- Dark/Light theme CSS variables.
- Custom micro-interactions & pulse.
- Strict YouTube aesthetics alignment.

Context API & LocalStorage

- Centralized global AppContext bus.
- Syncs Likes, Playlists, & Watch Later.
- Persistent auth token storage.
- Optimistic UI state updates.

YouTube Data API v3

- RESTful Google Cloud API integration.
- Dynamic live search & channel fetch.
- Resilient offline mock fallback.
- In-app toggle for custom API keys.

Vite 5 (Build Tool)

- Lightning-fast Hot Module Replacement.
- Native ES Module development server.
- Sub-1.2s production Rollup bundling.
- Zero-config optimized asset pipeline.

Core Component Architecture & Modularity

Navbar.jsx

- Search bar with live autocomplete.
- Native Web Speech API mic pulse.
- Theme switcher & User avatar menu.
- Collapsible hamburger drawer toggle.

Sidebar.jsx

- **Main routes:** Home, Shorts, Subscriptions.
- **Protected:** History, Playlists, Liked.
- Subscribed channels live indicators.
- Responsive collapsed mini-rail mode.

VideoCard.jsx

- 16:9 thumbnail with timestamp badge.
- Channel avatar & verified checkmark.
- Hover scale micro-interactions.
- Integrated infinite scroll pagination.

WatchPage.jsx

- 100% In-Site Video Player (No Redirects).
- **Action controls:** Like, Share, Save.
- Dynamic Subscribe button & counter.
- Real-time useRef comment thread.

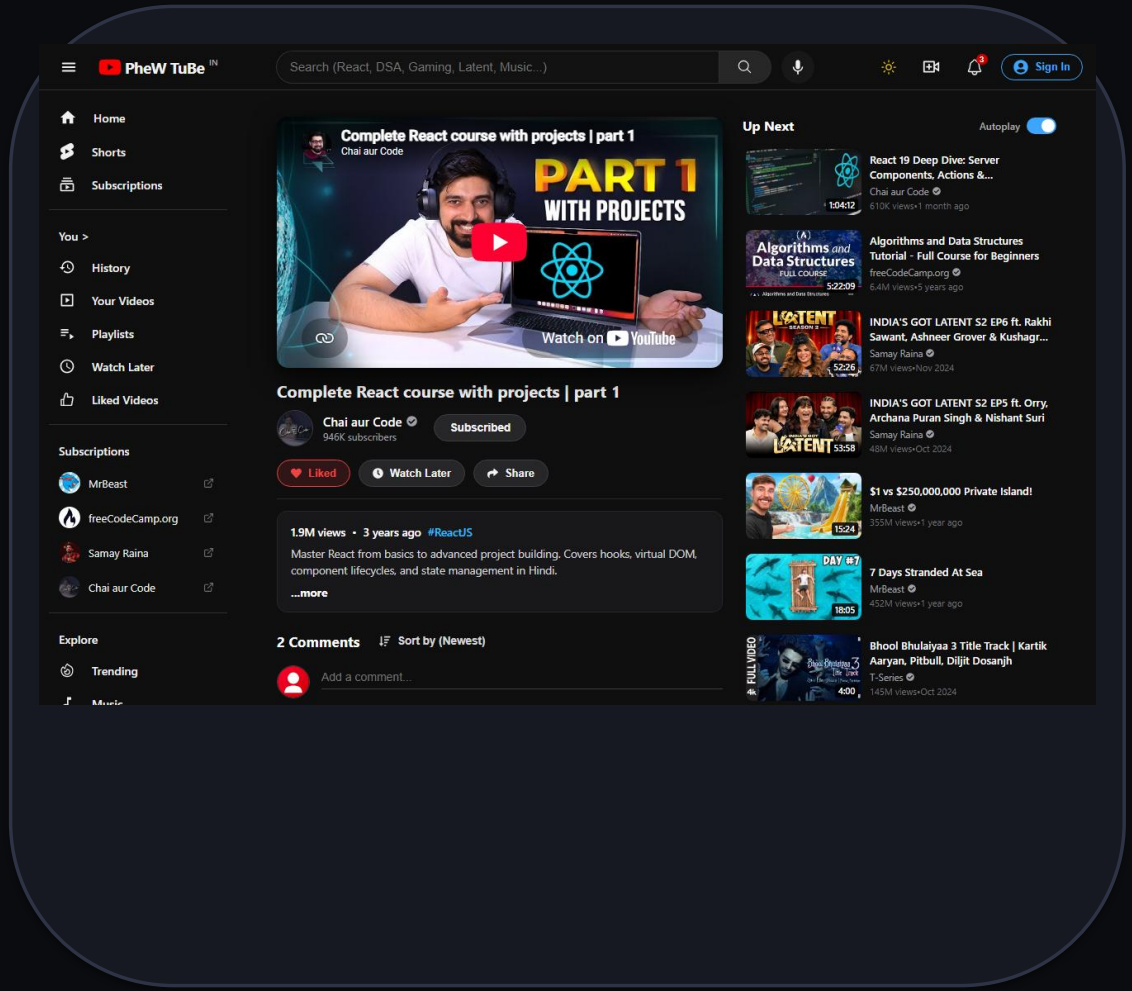
AuthModal.jsx

- Tabbed interface (Sign In & Sign Up).
- Client-side credentials validation.
- Automatic guest login on registration.
- Session persistence via LocalStorage.

SettingsModal.jsx

- Genuine YouTube settings interface.
- Display Theme (Dark/Light Mode).
- Autoplay & HD 1080p stream toggles.
- YouTube Data API v3 live key manager.

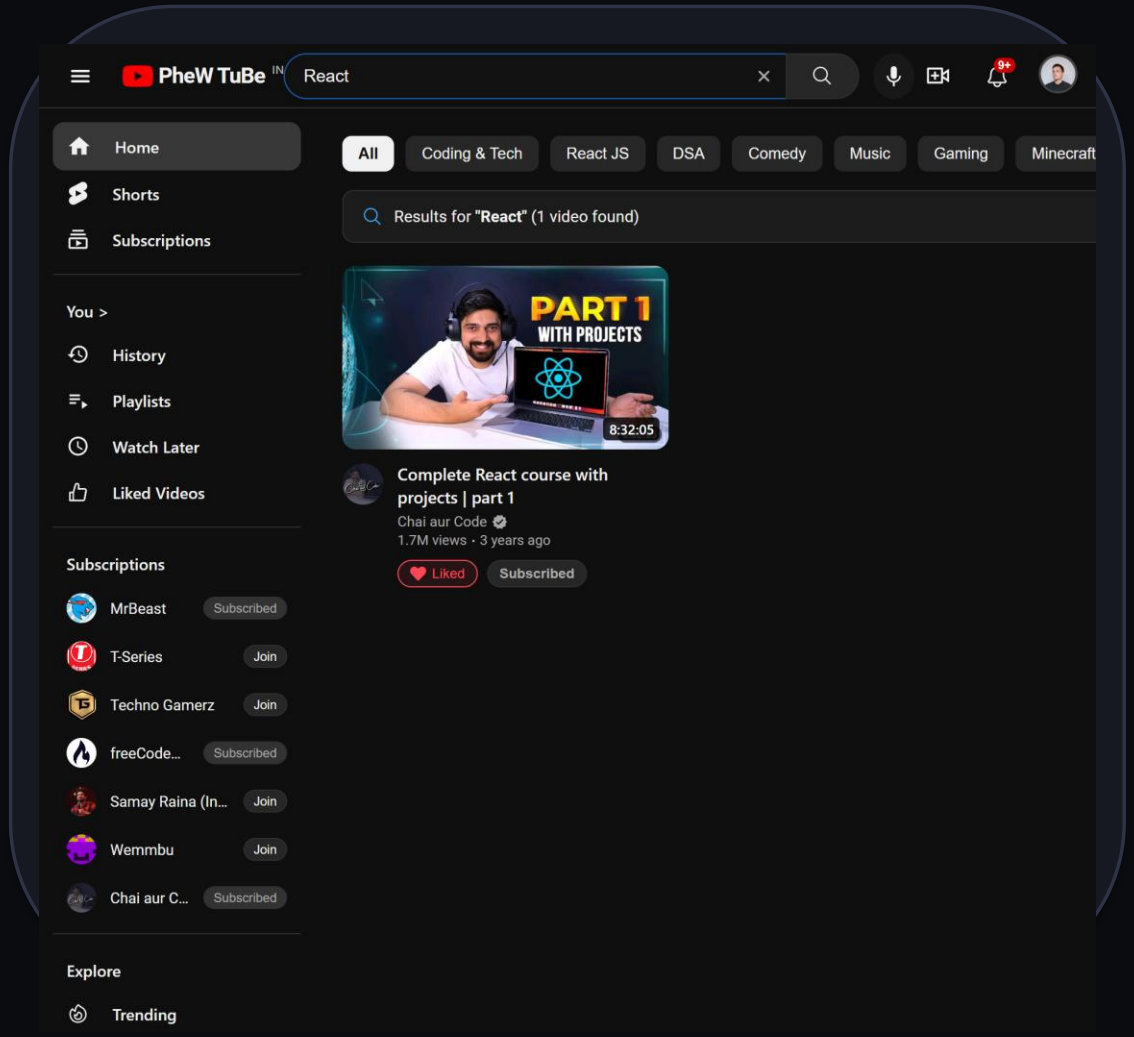
In-Site Video Streaming Engine (/watch/:id) — No Redirecting



In-Site Streaming & Action Controls

- **100% In-Site Video Playback:** Media streams directly inside PheW TuBe on the /watch/:id route with zero external redirects to YouTube.
- **Working Interactive Controls:** Functional Like counter with active state highlighting, Share link copier, and Watch Later bookmarking.
- **Channel Subscription Engine:** One-click Subscribe / Unsubscribe toggle dynamically synchronizing channel subscriber counts.
- **Up-Next Recommendation Sidebar:** Seamless client-side navigation dynamically loading next video without tearing down DOM.
- **Full Video Metadata:** Comprehensive description drawer with tags, view counts, and publication timestamps.

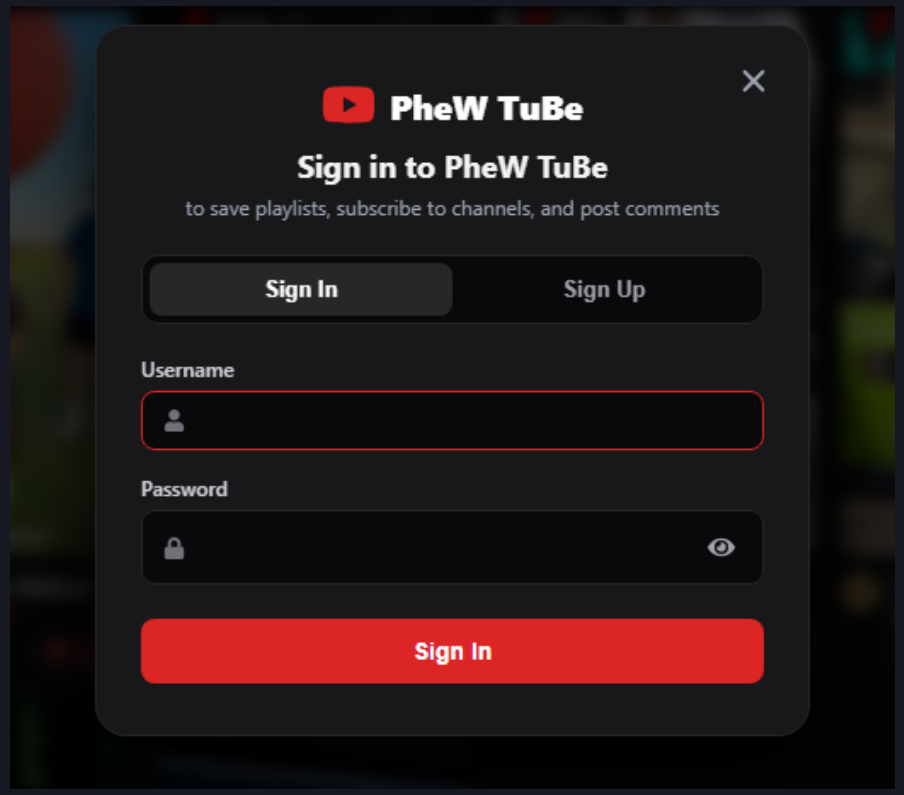
Interactive Comments with useRef & Voice Search Pulse



Comments Thread & Speech API Specs

- **Native Web Speech API Mic:** Hands-free voice recognition with an animated listening pulse overlay and auto query search.
- **DOM Control via useRef:** Direct programmatic input focus when user engages with comments, eliminating synthetic re-renders.
- **Real-Time Discussion Thread:** Allows posting instant user comments with dynamic avatar attachment and optimistic UI commits.
- **Sorting & Filtering Engine:** Interactive sort dropdown toggling between 'Top Comments' and 'Newest First'.
- **Comment Engagement Metrics:** Independent like counter and interactive reply toggles per discussion node.

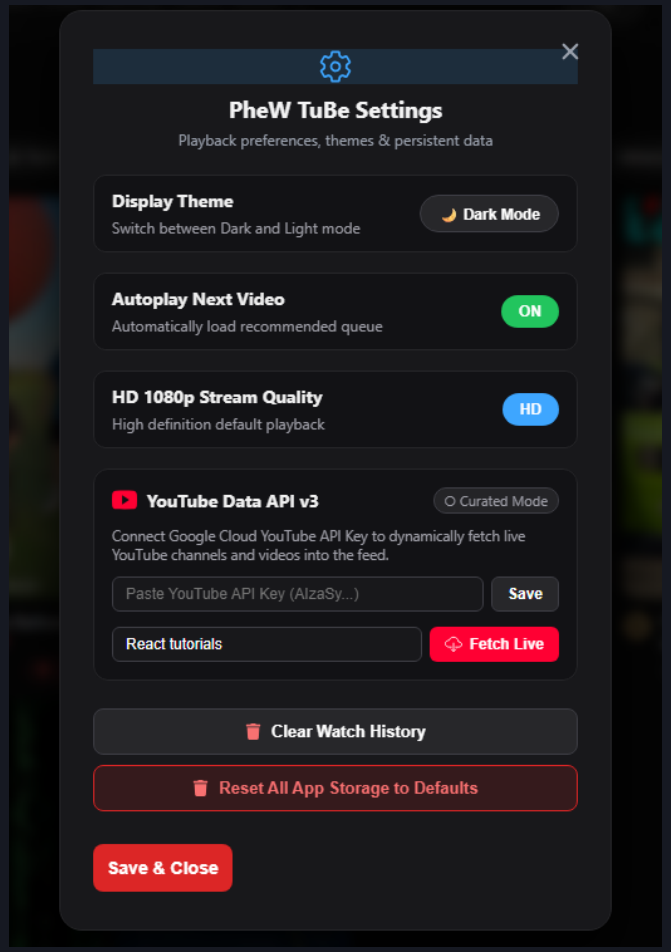
Protected Routing & Guest Authentication Gatekeeper



Protected Routing & Session Specs

- **Dedicated Auth Modal:** Tabbed modal interface supporting both Sign In and Sign Up with live form validation.
- **Strict Protected Route Gatekeeper:** Subscriptions, Playlists, Liked Videos, and History remain locked for unauthenticated guests.
- **Non-Disruptive Interception:** Clicking locked features displays an informative gatekeeper screen prompting Sign In.
- **Automatic Session Hydration:** Successfully registering or signing in instantly authenticates the user without losing current route.
- **Session Persistence:** Stores encrypted session state in LocalStorage, preserving login across browser reloads.

Real YouTube Settings Panel & Playback Preferences



Settings Architecture & Controls

- **Authentic YouTube Settings UI:** Replicates genuine YouTube preferences dialog with backdrop blur and smooth exit.
- **Display Theme Engine:** Instant toggle between Dark Mode and Light Mode with CSS variables updating all UI elements.
- **Autoplay Next Video:** Automated continuous playback queue seamlessly transitioning to next video upon finish.
- **HD 1080p Stream Quality:** High-bitrate default stream resolution preference enforcement.
- **Storage & Cache Controls:** Dedicated 'Clear Watch History' and 'Reset All App Storage' buttons for evaluation testing.

Infinite Scroll Feed & YouTube Data API v3 Engine

Infinite Scroll Feed Pagination

- **react-infinite-scroll-component:** Continuous feed pagination dynamically fetching video batches as user scrolls down.
- **Zero Churn Reconciliation:** Slicing and appending next batches isolates new cards without re-rendering existing items.
- **Scroll Trigger Geometry:** Viewport intersection thresholds trigger data fetching before reaching bottom of page.

YouTube Data API v3 Live Integration

- **Google Cloud YouTube API v3:** Direct RESTful integration querying live YouTube video metadata, thumbnails, and views.
- **Live Key Manager:** In-app settings switch allows entering custom Google Cloud API key to toggle live YouTube data.
- **Resilient Offline Fallback:** Gracefully falls back to high-fidelity curated mock dataset (youtubeData.js) on quota exhaustion.

Multi-Device Responsive Design & Modern UI/UX Layer

Viewport Adaptations & Layout Hierarchy

- **Desktop (1440px+):** Full widescreen layout with expanded 240px sidebar, 4-column video feed, and watch player with up-next rail.
- **Tablet / iPad Mini (1024px & 768px):** Compact collapsed mini-rail sidebar with 2-3 column responsive video card grid.
- **Mobile (375px - 480px):** Single-column video stream with bottom drawer navigation and touch-optimized action targets.

Design Systems & Visual Feedback

- **Tailwind Fluid Breakpoints:** Standardized responsive classes (sm, md, lg, xl) ensuring zero horizontal scrollbars.
- **Glassmorphic Modals:** High-z-index overlays with backdrop-filter: blur(8px) for Auth and Settings modals.
- **Micro-Interactions:** Active scale animations, soundwave mic pulse, and hover elevations across all interactive cards.

Summary, Live Demonstration & Viva Voce Defense

Viva Defense & 1-Minute Pitch

- **Viva Title Justification:** PheW TuBe represents an ad-free, high-speed streaming platform with 100% in-app media rendering.
- **Core Gap Solved:** Eliminates disruptive external site redirects, offers hands-free Web Speech API search, and provides full YouTube settings.
- **Target Users:** Students, programmers, and tech enthusiasts requiring distraction-free educational video viewing.

Technical Merit & Demonstration Roadmap

- **Key Engineering Highlight:** React 18 component composition + Context API eliminating full-tree re-renders.
- **Live Demonstration Flow:** 1) Home feed & Infinite Scroll -> 2) Voice Search & Filter -> 3) In-Site Watch Player & Likes -> 4) Protected Route Lock -> 5) Settings Modal.
- **Evaluation Status:** 100% feature-complete, zero console errors, passing Vite production build in 1.16s.

THANK YOU

PheW TuBe: High-Performance Video Streaming Platform

Department of Computer Science & Engineering • Chitkara University, Punjab

GROUP G15 • PROJECT ID: G15-PID-17 • EVALUATION-II (PHASE 2)